THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. $M(-1, 4)$ is the midpoint of AB and $A(5, 6)$. The coordinates of B are: 1

(a) $(4, 10)$ **(b)** $(-7, -2)$ **(c)** $(-6, -2)$ **(d)** $(-7, 2)$

2. The distance between $A(-2, 3)$ and $B(3, -9)$ is: 1

(a) 14 units **(b)** 15 units **(c)** 12 units **(d)** 13 units

3. A linear growth is represented by a straight line with: 1

(a) undefined slope **(b)** positive slope **(c)** negative slope **(d)** zero intercept

4. The value of $(-20) \div 4$ is: 1

(a) 4 **(b)** -5 **(c)** 5 **(d)** -3

5. The coefficient of ab in $(a + b + c)^2$ is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

6. A chord of length 48 cm is at a distance of 7 cm from the centre of a circle. The radius of the circle is: 1

(a) 25 cm **(b)** 26 cm **(c)** 48 cm **(d)** 31 cm

7. The area of a parallelogram with base 15 cm and height 8 cm is: 1

(a) 60 cm^2 **(b)** 46 cm^2 **(c)** 120 cm^2 **(d)** 135 cm^2

8. If $P(E) = 0.35$, then the probability that E does not happen is: 1

(a) 0.65 **(b)** 0.55 **(c)** 0.35 **(d)** 1.35

9. The 7th term of the GP with first term 1 and common ratio -2 is: 1

(a) 63 **(b)** 64 **(c)** 65 **(d)** 128

10. Which of the following is a counterexample to the proposition "If a number is odd, then it is prime."? 1

(a) 7 **(b)** 5 **(c)** 9 **(d)** 11

11. **Assertion (A):** The number 12 is a counterexample to "If a number is a multiple of 3, then it is a multiple of 9". 1

Reason (R): 12 is a multiple of 3 but not a multiple of 9.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

12. A group of 40 students has an average of 60 marks and another group of 10 students has an average of 85 marks. The average of all 50 students is: 1

(a) 64 **(b)** 66 **(c)** 65 **(d)** 67

13. **Assertion (A):** The simplest gcd algorithm that scans every j from 1 to min(m, n) takes 10 times as much work when the smaller number goes from 3 digits to 4 digits. 1
Reason (R): The work of that algorithm is proportional to the digits of the numbers.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

14. At the start of the algorithm for the divisors of n, the list-of-divisors is: 1

(a) [1] **(b)** [n] **(c)** [1, n] **(d)** empty

15. The three medians of a triangle meet at a point called the: 1

(a) centroid **(b)** incentre **(c)** circumcentre **(d)** orthocentre

16. The converse of 'the opposite angles of a parallelogram are equal' is: 1

(a) all quadrilaterals have equal opposite angles
(b) if the opposite angles of a quadrilateral are equal, it is a parallelogram
(c) if a quadrilateral has equal angles, it is a rectangle
(d) if a quadrilateral is a rectangle, opposite angles are equal

17. The solution of the pair 2x + 3y = 5 and x − 2y = 6 is: 1

(a) (-1, 4) **(b)** (4, -1) **(c)** (-4, -1) **(d)** (4, 1)

18. The graph of a linear equation in two variables is: 1

(a) a single point **(b)** a circle **(c)** a straight line **(d)** a parabola

19. Taking $\pi = \frac{22}{7}$, the curved surface area of a hemisphere of radius 7 cm is: 1

(a) 462 cm^2 **(b)** 154 cm^2 **(c)** 308 cm^2 **(d)** 616 cm^2

20. The volume of a cube is 64 cm^3. Its total surface area is: 1

(a) 48 cm^2 **(b)** 96 cm^2 **(c)** 16 cm^2 **(d)** 64 cm^2

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. In which quadrant (or on which axis) do the points $(-2, 5)$, $(6, -3)$, $(0, -4)$ and $(-1, -1)$ lie? 2

22. Locate $\frac{3}{4}$ on the number line by describing the steps. 2

23. In a circle of radius 5 cm, one chord is 3 cm from the centre and another is 4 cm from the centre. Which is longer? Find both lengths. 2

24. The perimeters of two circles are in the ratio 5 : 4. What is the ratio of their radii? 2

25. How many factors does 64 have? Is the number of factors odd or even? What does this say about 64? 2

26. Compute gcd(60, 16) by the long-division (division) algorithm. 2

27. True or false, with a reason: a parallelogram with one right angle is a rectangle. 2

28. A tank is 2 m long, 1 m wide and 1.5 m deep. How many litres of water can it hold? (1 $\text{m}^3 = 1000$ litres) 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Show that the sum of the angles of a convex quadrilateral is 360°. 3

30. A drink needs 8% fruit pulp. How many litres of 20% pulp concentrate should be mixed with 6 litres of 5% pulp drink to get 8%? 3

31. The volume of a box of length 7 cm and breadth 11 cm is a linear polynomial in the height h. Find the volume for $h = 5, 9, 13$. 3

32. In cyclic quadrilateral ABCD, $\angle A = (4x - 4)^\circ$ and $\angle C = (2x + 16)^\circ$. Find x and both angles. 3

33. Find a point on the y-axis equidistant from $A(6,\ 5)$ and $B(-4,\ 3)$. 3

34. Use the converse of the Baudhāyana–Pythagoras theorem to find which of these triples can be the sides of a right triangle: 3
(i) 8, 15, 17
(ii) 10, 24, 27.

35. A die is rolled once. Find the probability that the number is both even and prime, and the probability that it is neither. 3

OR

Distinguish between subjective and objective (experimental/theoretical) probability using the example ‘it will rain today’. 3

36. The edge of a cube is increased by 50%. Find the percentage increase in 3
(i) its total surface area,
(ii) its volume.

OR

A hemispherical bowl has an inner radius of 10.5 cm. Find the volume of liquid it can hold in litres ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Prove the Centroid Theorem: the medians of a triangle are concurrent and divide each other in the ratio 2 : 1. (Use midpoints of CM and BM.) 5

OR

Prove that the midpoints of the sides of a quadrilateral ABCD form a parallelogram. If the diagonals are AC $= 16$ cm and BD $= 16$ cm, name the parallelogram and find its perimeter. 5

38. A boat goes 20 km downstream in 2 hours and 12 km upstream in 2 hours. Find the speed of the boat in still water and the speed of the stream. 5

OR

The sum of the length and the width of a rectangle is 22 cm. If the length is decreased by 2 cm and the width is increased by 3 cm, the rectangle becomes a square. Find the original dimensions and the area of the rectangle. 5

39. Classify each as AP, GP or neither, and give the nth term where it exists: 5
(a) $7, 11, 15, 19, \ldots$
(b) $2, 6, 18, 54, \ldots$
(c) $1, 4, 9, 16, \ldots$
(d) $100, 50, 25, 12.5, \ldots$

OR

Case study: Meera saves ₹100 in the first month and ₹50 more each month than in the previous month. 5

(i) Write her savings for the first three months.

(ii) Find the nth term.

(iii) Find her saving in the 12th month and her total in a year.

(iv) In which month does she save ₹1,000?

40. **Case study:** Saira arranges 1 square of side x, 8 rectangular strips of $x \times 1$ and 15 unit squares into a rectangle. A rectangular pool has breadth 4 m less than its length and area 96 m². 5

(i) Find Saira's rectangle's length and breadth.

(ii) Form and solve the equation for the pool.

(iii) Find the pool's perimeter.

OR

The sum of three numbers is 10, the sum of their squares is 38 and their product is 25. 5

(i) Find $xy + yz + zx$.

(ii) Hence find the sum of their cubes.

END OF QUESTION PAPER **Total: 80 Marks**

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